

Chapter 108 — Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **POP-Q** (Pelvic Organ Prolapse Quantification) = **recommended staging**; hymen = 0, cm, stage by most distal point
- **Stage 2** = -1 to +1; **stage 3** > +1 to < +(tvI - 2); **stage 4** ≥ +(tvI - 2)
- **Asymptomatic stage 1-2 POP (pelvic organ prolapse)** = **normal**; symptoms start at the hymen
- **Most specific POP symptom**: seen or felt vaginal bulge
- **Anterior commonest**; apical almost always present with significant anterior POP → **add vault suspension**
- **CS (caesarean section) NNT (number needed to treat) 13** to prevent 1 moderate-severe UI (urinary incontinence); lost with any vaginal birth
- **OPUS**: prophylactic MUS (mid-urethral sling) at vaginal POP repair → UI 27.3% vs 43%, **NNT 6.3**, more harms
- **VLPP (Valsalva leak point pressure) <60 cm H₂O** → **ISD** (intrinsic sphincteric deficiency)
- **Post-RP (radical prostatectomy) UI**: 8-60% at 1 year; **ISD >70%** on UDS (urodynamic studies)
- **Male remission up to 40%** vs female ≈12%/year; male UUI (urgency UI) > MUI (mixed) > SUI (stress)

DEFINITION AND CLASSIFICATION OF UI

- **Treat for bother**: QoL (quality-of-life) impact = commonest reason to intervene
- **Sign** = clinician-observed; **symptom** = patient report; **observation** = diary or UDS

TERMINOLOGY OF LUTS AND INCONTINENCE

- **Standard**: ICS/IUGA (International Continence Society / International Urogynecological Association) joint report, 2010
- **Frequency**: ≥8 voids/24 h
- **UD (urodynamic) SUI**: leak with ↑ abdominal pressure, **no detrusor contraction**
- **UUI needs no DO** (detrusor overactivity) on UDS
- **Continuous UI** → **fistula** (vesicovaginal, ureterovaginal)
- **Increased sensation**: early desire, still deferrable; may be <50 mL filling

EPIDEMIOLOGY OF UI IN FEMALES

- **UI in pregnancy ≈40%**; SUI ≈two-thirds
- **Primiparous**: ≈3× UI in pregnancy; ≈2.5× at 1 year
- **Hysterectomy resolves UI in 13%**: uterus >300 g, BMI (body mass index) <25, coexisting POP
- **Vaginal/laparoscopic hysterectomy** → more UI than open
- **Annual incidence**: 1-10% at 40-60 (≈3% per subtype); 10-20% older
- **Annual remission** up to ≈12%; **MUI remits least**
- **Menopause (48-54)** → ↑ SUI incidence, not UUI

RISK FACTORS FOR UI IN FEMALES

- **Age**: strongest non-modifiable; SUI peaks 40-59 → UUI/MUI >60
- **LTCF** (long-term care facility) UI >40%; **ADL (activities of daily living) impairment** strongest link
- **Nonagenarians**: UI = independent risk factor for death
- **Vaginal delivery** → **SUI, not UUI**; baby ≥4 kg → OR ≈1.5
- **Single delivery** → OR ≈1.5; strongest for SUI
- **First pregnancy >30** → OR ≈3 (vs other data: near 20 worst)
- **BMI ≥30** → >2× UI; reversible with weight loss
- **Oral oestrogen ± progestogen** → ≈2× incident UI; vaginal oestrogen not
- **White > Black/Asian** (↑ urethral closure pressure in Black women)
- **Caffeine** → OAB (overactive bladder)/UUI, not SUI; **smoking** → urgency
- **Type 2 DM (diabetes mellitus)** → up to 2× UI; macroalbuminuria, neuropathic pain

EPIDEMIOLOGY OF UI IN MALES

- **Prevalence 11-34%** older men; up to 11% daily
- **Incidence ≈1-10%/year** at 60-70
- **Male SUI** = radical pelvic surgery or neurological disease
- **TURP** (transurethral resection of the prostate) UI ≈1%
- **Robotic vs open RP**: comparable UI

DEFINITIONS AND CLASSIFICATION OF POP

- **Name the segment** (anterior, apical, posterior), not cystocele/rectocele
- **Baden-Walker**: halfway system, grades 0-4, hymen reference, high interobserver variability
- **POP-Q**: developed 1993, published 1996 (Bump), updated 2002, recommended 2016
- **Aa/Bp**: 3 cm inside hymen; range -3 to +3
- **Ba/Bp** = -3 normal; = C in total eversion
- **D omitted** without cervix; **gh, pb, tvI** measured at rest
- **Stage 1**: >1 cm above hymen (< -1)
- **Simplified POP-Q**: 4 points (Ba, Bp, C, D)
- **Postmenopausal bleeding** never from POP → gynaecological evaluation
- **Pessary trial** positive → surgery likely to help
- **Imaging** usually unnecessary; MRI (magnetic resonance imaging) defecography for enterocele or mismatch

EPIDEMIOLOGY OF POP

- **Symptomatic POP 3-12%**; support loss on examination 43-90%
- **POSST** (Pelvic Organ Support Study): no POP 24%; stage 1 38%, stage 2 35%, stage 3 2%
- **WHI** (Women's Health Initiative): anterior 34%, posterior 19%, uterine 14%
- **WHI incidence/100 woman-years**: cystocele 9.3, rectocele 5.7, uterine 1.5
- **Progression**: age >60, Ba ≥4 cm, gh ≥5 cm
- **Oxford Family Planning Study**: RR (relative risk) 8.4 (2 births), 10.85 (≥4)
- **Each extra delivery** (up to 5) → +10-20%
- **Forceps** > vaginal > CS only > nulliparous
- **Labored CS ≈ vaginal**; elective CS ≈ nulliparous
- **Age**: ≈+40% prevalence per decade; surgery peaks in 7th decade
- **Hysterectomy for POP** → 5.5× POP; 29 vs 16/1000 woman-years
- **Constipation** → 2.5×; sibling of severe POP ≈5×

- **Hispanic highest, Black lowest;** oestrogen status not consistently linked
- **Recurrence:** preoperative stage 3–4

RELATIONSHIP BETWEEN UI AND POP

- **SUI → anterior POP in 40–55%;** OAB → anterior POP ≈20%
- **OAB improves in 60–80%** after POP repair; anterior/apical > posterior
- **Occult SUI:** seen only after POP reduced; up to ¼ of advanced POP
- **De novo SUI risk:** age >66, DM, low urethral closure pressure
- **TOMUS** (Trial of Mid-Urethral Slings): stage 2 POP 49%, stage 3 6%
- **CARE** (Colpopexy and Urinary Reduction Efforts): Burch at sacrocolpopexy → postoperative SUI 38% vs 58%
- **OPUS** harms: perforation 6.7%, UTI (urinary tract infection) 31% vs 18.3%, bleeding 3.1%

CONSEQUENCES OF UI AND PROLAPSE

- **UI costs >\$50 billion/year** (United States); female SUI >\$12 billion, >70% out-of-pocket
- **OAB:** ≈42 million Americans, ≈\$25 billion/year
- **Most bothersome LUTS** (lower urinary tract symptoms): urgency and UUI
- **Lifetime surgery risk** for POP/UI 11% (Olsen 1997); reoperation 29%
- **POP** = commonest hysterectomy indication >54
- **38% with POP** avoid intercourse

PHYSIOLOGY OF URINARY CONTINENCE

- **Storage:** parasympathetic (pelvic) off; sympathetic (hypogastric) on
- **Onuf's nucleus:** sacral pudendal motor nucleus → tonic external sphincter
- **Compliant bladder:** type I collagen; poor detrusor cell coupling
- **Rhabdosphincter:** type I slow-twitch; >40 cm H₂O continuously
- **Male bladder neck** competent; female often incompetent
- **Female mucosal seal:** oestrogen-sensitive; up to 30% of continence force

PATHOPHYSIOLOGY OF UI: GENERAL PRINCIPLES

- **MSA** (multiple system atrophy): ≥20% UI after TURP
- **RP** = commonest surgical cause of male UI
- **Suprapontine lesion** (CVA (cerebrovascular accident), MS (multiple sclerosis), Parkinson disease) → NDO (neurogenic DO) → UUI
- **Conus/lumbosacral lesion**, neuropathy → overflow
- **Cervical/upper thoracic SCI** (spinal cord injury) → DSD (detrusor–sphincter dyssynergia) + NDO

PATHOPHYSIOLOGY OF SUI IN FEMALES

- **Pressure transmission:** Enhorning 1961 — urethral descent out of abdominal pressure zone

- **Hammock:** DeLancey 1994 — urethra compressed against firm vaginal backing
- **Integral theory:** Petros and Ulmsten 1990 — PUL (pubourethral ligament) laxity → basis of MUS
- **ISD:** McGuire and Lytton 1978 — pubovaginal fascial sling
- **Type III UI** = ISD (Blaivas and Olsson 1988); "pipestem" urethra
- **Burch** is less effective in severe ISD

PATHOPHYSIOLOGY OF INSENSIBLE INCONTINENCE

- **Post-void loss** → diverticulum → MRI; **continuous** → ectopic ureter → upper tracts

PELVIC FLOOR SUPPORT MECHANISM

- **DeLancey level I** (upper third; uterosacral–cardinal) → apical prolapse
- **Level II** (mid; ATRP (arcus tendineus fasciae pelvis)) → anterior wall prolapse
- **Level III** (distal; perineal membrane) → urethral hypermobility, distal rectocele, perineal descent
- **Richardson 1981:** commonest cystocele defect = ATRP detached from ischial spine

PATHOPHYSIOLOGY OF POP

- **Levator defects** on MRI in up to 20% of primiparous women, 0% of nulliparous
- **Levator nerves S3–S5;** pudendal nerve → sphincters and perineum
- **Ehlers–Danlos/Marfan** → ↑ POP

CLASSIC TRAPS

- **Trap:** UUI needs DO on UDS — Actually: no UD finding required
- **Trap:** CS always protects — Actually: lost with one vaginal birth; labored CS ≈ vaginal for POP
- **Trap:** Isolated cystocele → anterior repair alone — Actually: apex almost always involved
- **Trap:** Constipation → posterior repair — Actually: only if bulge or splinting
- **Trap:** POP causes pain or bleeding — Actually: look for other causes
- **Trap:** Vaginal oestrogen causes UI — Actually: oral does; vaginal does not
- **Trap:** Stage 1–2 POP needs repair — Actually: normal if asymptomatic
- **Trap:** Hammock = reposition urethra — Actually: restore backing
- **Trap:** Burch works in ISD — Actually: sling/bulking; Burch fails
- **Trap:** Pessary removed → measure — Actually: underestimates prolapse